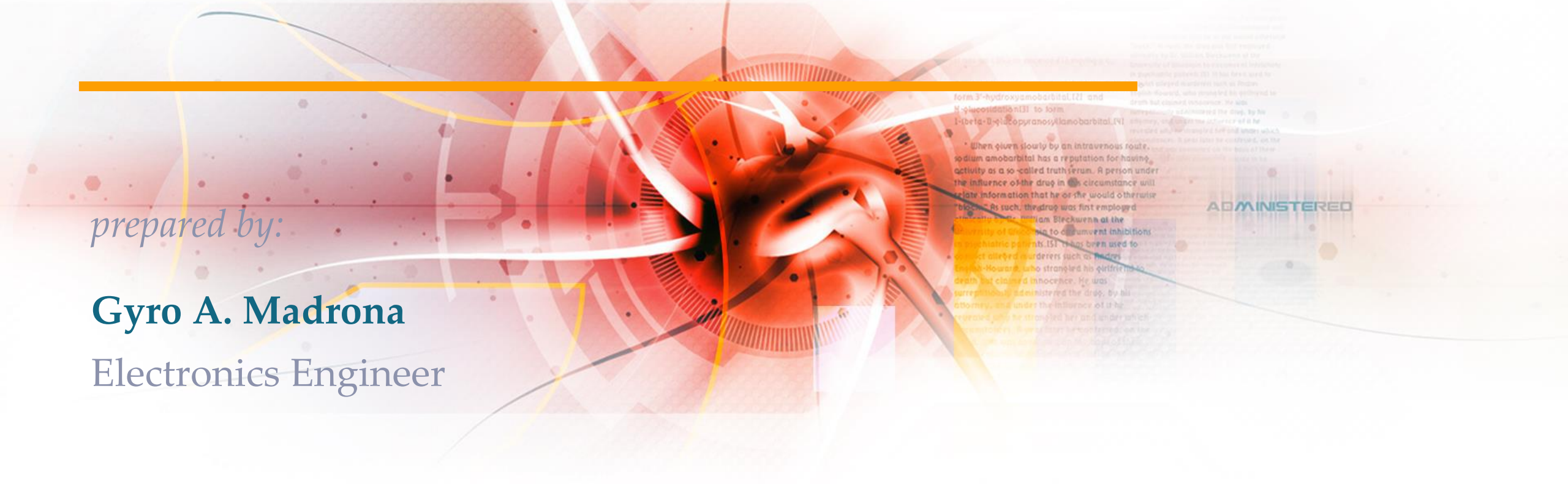
PRACTICAL APPLICATIONS

SPHERICAL TRIGONOMETRY



prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

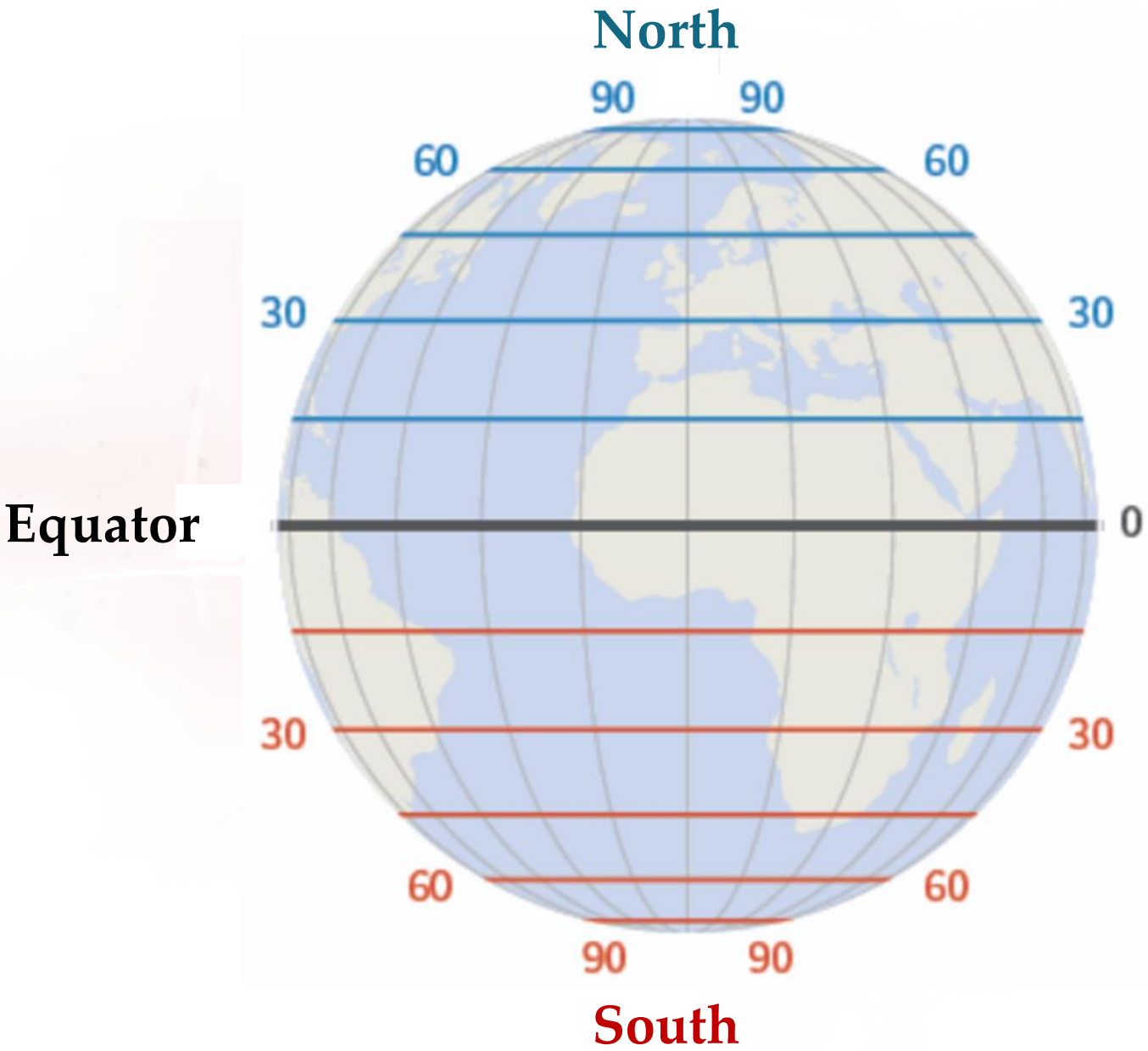
Latitude and Longitude

Time Difference

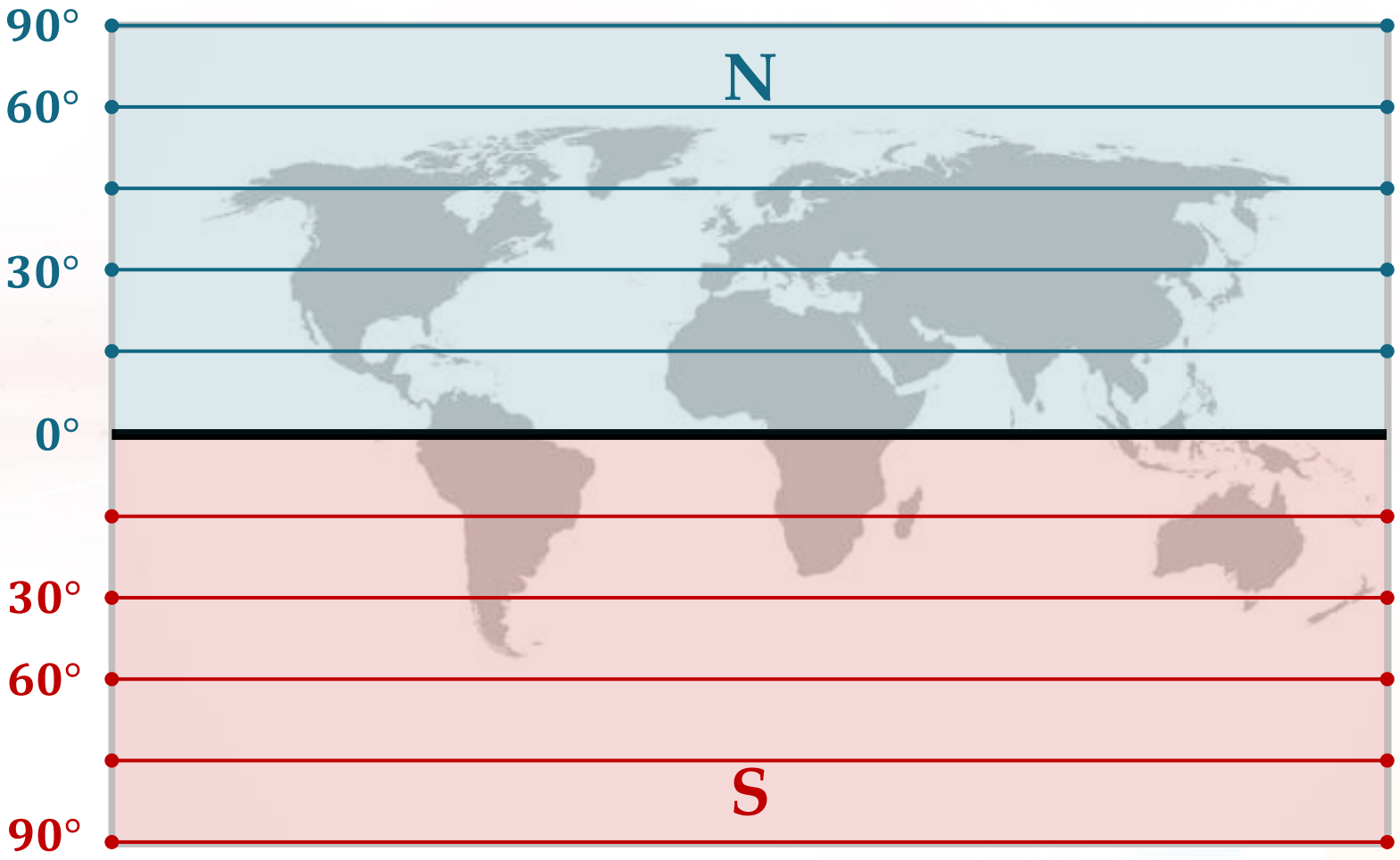
Distance Between Two Locations

LATITUDE AND LONGITUDE

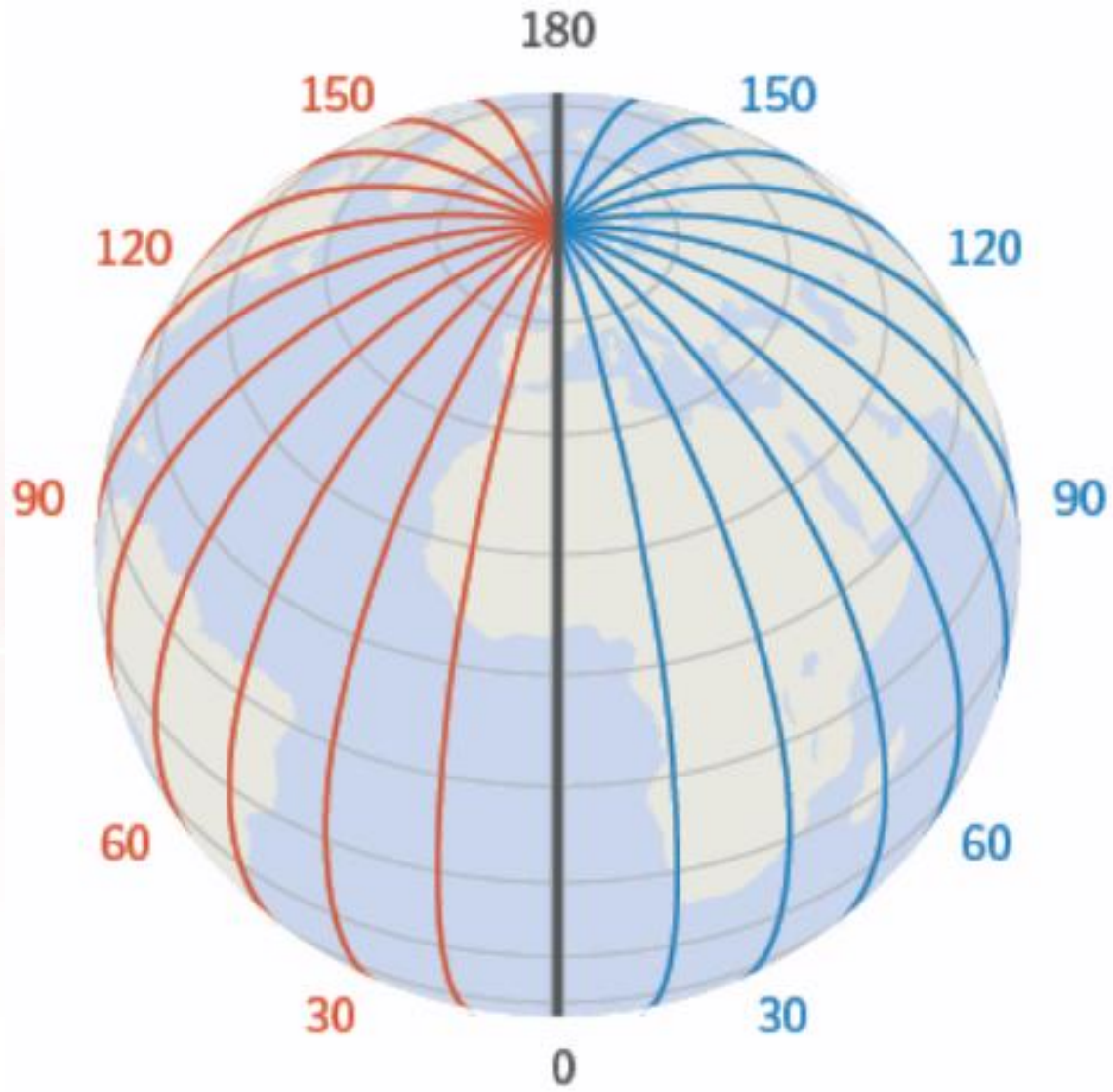
LATITUDE



EQUATOR



LONGITUDE

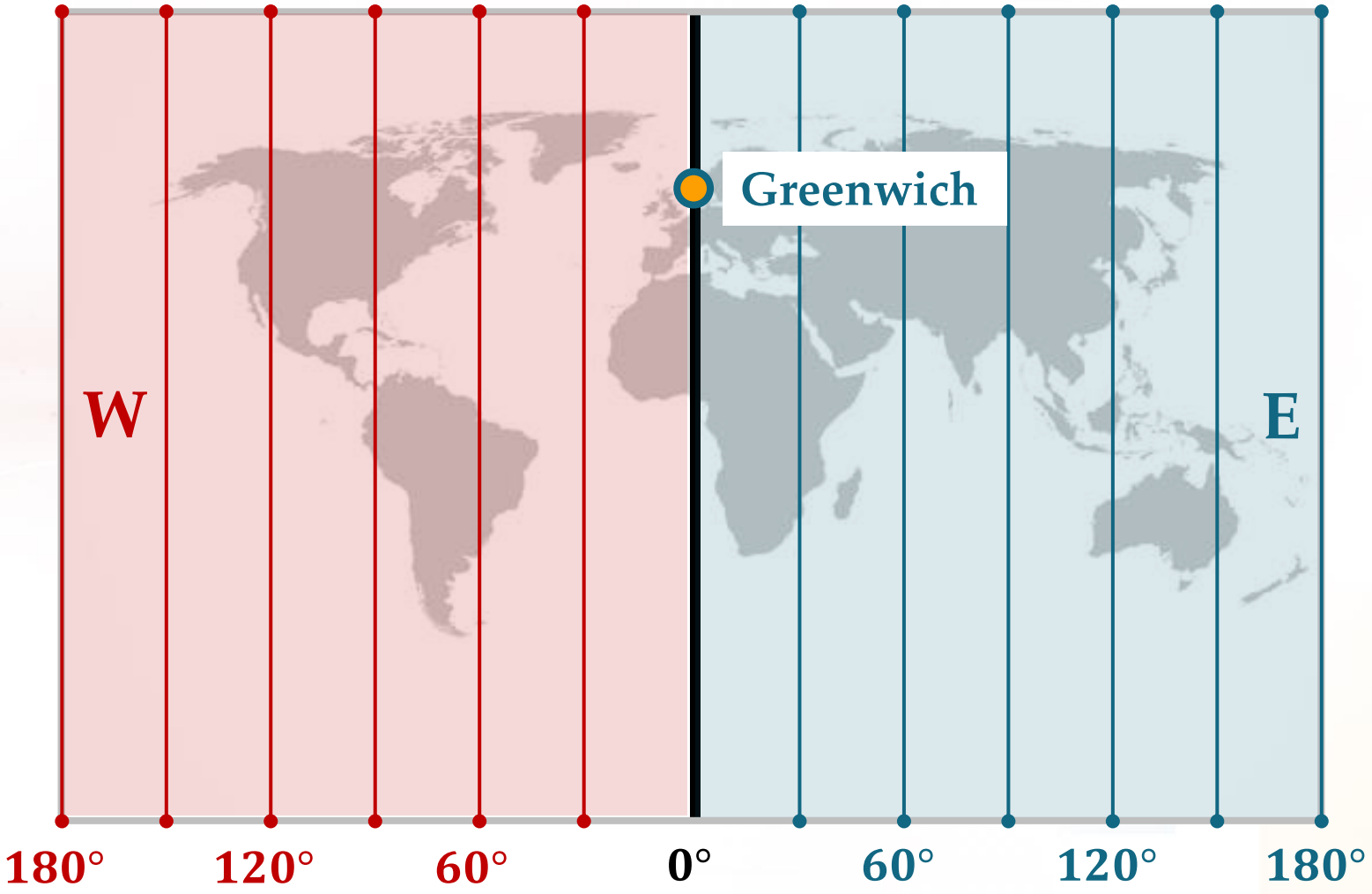


West

East

Prime Meridian

PRIME MERIDIAN



TIME DIFFERENCE

EXERCISE

If the longitude of Tokyo is $\mathbf{139°}\boldsymbol{E}$ and that of Manila is $\mathbf{121°}\boldsymbol{E}$, what is the **time difference** between Tokyo and Manila?

EXERCISE

If the time is **8: 00 AM GMT**, what is the **time** in the **Philippines**, which is located at $\mathbf{120°}\boldsymbol{E}$ **longitude**?

DISTANCE BETWEEN TWO LOCATIONS

EXERCISE

Tokyo is located at $\mathbf{139°E, 39°N}$ while Manila is at $\mathbf{121°E, 14°N}$. Find the **distance** between the two in nautical miles.

SEATWORK